

Vanishing Volumes: Constraining Reconstructions of Fine-Structured Objects from Coarse Segmentation

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Abstract

Recent advances at the interface of machine learning and 3D reconstruction, ranging from volumetric methods like NeRF and Plenoxels to point-based primitives such as Gaussian Splats, have enabled high-fidelity 3D modeling from images. Despite this progress, capturing assets through the isolated optimization of segmented objects remains fundamentally under-constrained. Additionally, thin structures pose a difficult challenge for segmentation methods. In this work, we explore necessary constraints and combine priors from monocular depth as well as visual hulls to overcome their respective failure modes, producing high-quality object-centric reconstructions in the face of erroneous segmentation. This is demonstrated using synthetic scenes, all exhibiting fine structure, which we openly release along with coarse and ground truth segmentation masks. Furthermore, we show that segmentation failure can act as a useful signal to guide sampling and further enhance detail preservation.

CCS Concepts

• **Computing methodologies** → **Reconstruction; Volumetric models; Image segmentation;**

1. Introduction

Modern 3D reconstruction research has produced a diverse set of representations, spanning volumetric approaches such as neural radiance fields (NeRFs) [MST*20], faster, voxel-based radiance fields like plenoxels [YFT*21], and point-based techniques such as 3D Gaussian Splatting (3DGS) [KKLD23]. These methods are capable of modeling complex geometry and appearance from 2D observations, producing convincing novel view synthesis (NVS) and ultimately help bridge the gap between traditionally captured media and computer graphics pipelines.

To facilitate editing, such representations may be decomposed into discrete objects that can be manipulated and stored as modular assets. Objects are often captured using segmented training images, drawing on the vast amount of research in the 2D domain. Despite significant progress in 2D segmentation, current models still struggle to capture fine details like hair strands or intricate foliage, especially within complex, cluttered scenes. These conditions can result in missed regions as well as unwanted scene elements within predicted masks. For high-fidelity applications, such errors can be detrimental. More fundamentally, optimizing single objects is highly under-constrained and prone to nonsensical reconstruction, despite being considerably more efficient than full-scene training. To address both issues, we propose a framework that combines monocular depth estimation and visual hull-guided sampling; we demonstrate that each strategy helps combat the other's failure modes and that together, they provide strong constraints which enable efficient, high-quality reconstruction of isolated objects, even in the face of segmentation errors. We adopt voxel-based radiance

fields as our base representation as they build truly volumetric, photo-realistic representations in minutes. Additionally, their regular structure makes it straightforward to define discrete objects and extract mesh assets through methods like marching cubes [LC87]. Our key contributions are summarised below:

- A robust framework for reconstructing fine-structured objects with high fidelity from coarse segmentation across viewpoints.
- A novel strategy which exploits segmentation failure to identify and help prioritize fine structure.
- We release a multi-scene synthetic benchmark featuring objects with thin structure, including coarse and ground-truth segmentation to facilitate evaluation of reconstruction methods under imperfect segmentation.

2. Related Work

2.1. Object-Centric Representations

Motivated by applications like scene editing and mixed reality, several prior works have proposed methods for capturing standalone radiance field assets. A prominent, early example is ObjectNeRF [YZX*21] which adopts two separate branches for scene and object rendering. Later, DM-NeRF [WCY22] proposes a method that learns spatial codes for individual 3D objects using 2D supervision, isolating them for transforms like translation and rotation. While direct media applications of such technology are clear (some of which are explored in [BCE*24]), elements of object-centricity have also been adopted within wider pipelines. For example, TI-

Face [ZCS*23] builds on tensorial radiance fields, using 2D segmentation to isolate faces from backgrounds. While object-centric methods have grown considerably, the majority of approaches incorporate full-scene information and often assume perfect segmentation. Ideally, capturing a discrete asset should be robust to such errors and, for efficiency, ignore background pixels to not waste compute. Addressing both challenges would greatly enhance compositional scene representations.

2.2. Incorporating Extra Constraints

Foreground-only capture and segmentation error introduce reconstruction ambiguities that require additional constraints. Generic measures, like sparsity regularisation, can help tackle floating density in ambiguous scenarios but are not tailored to the fundamental problems in question. Visual hulls have been used as a stronger constraint to guide scene sampling [KIT*21], therefore restricting density placement and speeding up reconstruction. This work acknowledges intrinsic visual hull limitations, however, like the inability to model concavities and over-estimation in real-world settings. Parallel research has also turned to depth-based priors, although these strategies remain largely disparate. A notable example is DaRF [SPA*23] which uses monocular depth estimates for robust geometry reconstruction and artifact removal. Despite encouraging results, in this paper we show that depth alignment can be cheated in this specific, object-centric context.

3. Method

Our methodology (summarised in Fig. 2) applies complementary constraints to reconstruct objects given weak supervision: coarse segmentation and no sampling of background regions. While monocular depth estimates provide geometric priors, we demonstrate how radiance fields can “cheat” alignment. Visual hulls, which are themselves notorious over-estimators, help overcome depth ambiguity by defining a *minimum* distance between the camera and scene structure. Both coarse and refined masks are then used to identify fine structure for biased sampling. Key details and additional steps are described in the following subsections.

3.1. Base Representation

Feature Grids. Grid-based radiance fields are volumetric representations, formed by discretizing space into a regular grid structure, storing learned features at vertices. Our implementation closely follows Plenoxels, which models scenes as grids of volumetric opacity, σ , and view-dependent colour, $\mathbf{c}$, in the form of spherical harmonic coefficients (9 per color channel). These features are queried using trilinear interpolation and then volume rendered into images. For all scenes, our implementation used grids of size 500^3 .

Volume Rendering. Volumetric methods like NeRFs and Plenoxels typically assume simple emission-absorption light transport:

$$\hat{\mathbf{C}}(\mathbf{x}) = \int_{t_n}^{t_f} V(t) \sigma(\mathbf{x}(t)) \mathbf{c}(\mathbf{x}(t), \hat{\mathbf{d}}) dt. \quad (1)$$

$\hat{\mathbf{C}}$ is the colour of a camera ray $\mathbf{x}(t) = \mathbf{x}_0 + t\hat{\mathbf{d}}$ with origin $\mathbf{x}_0$ and direction $\hat{\mathbf{d}}$. $\mathbf{c}(\mathbf{x}(t), \hat{\mathbf{d}})$ is the color at a 3D point as seen from $\hat{\mathbf{d}}$

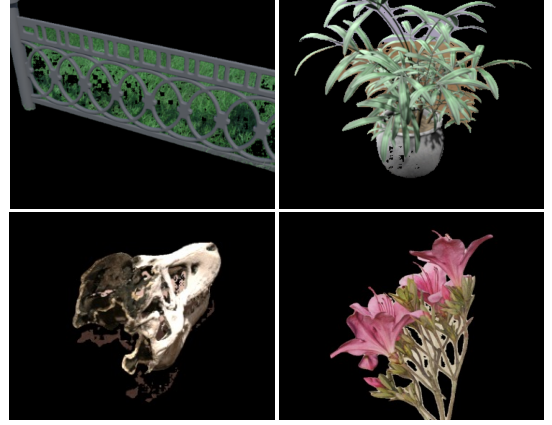


Figure 1: One example of segmentation failure from each of our four synthetic scenes.

and $V(t) = e^{-\int_0^t \sigma(\mathbf{x}(s)) ds}$ represents the decay of intensity as light passes through a density field. $V(t)\sigma(\mathbf{x}(t))$ behaves like an unnormalized probability density so equation (1) can be viewed as the color expectation along a ray. A mean squared error (MSE) loss then encourages rendered pixel colors, $\hat{\mathbf{C}}$, to match those within target images, $\mathbf{C}$: $\mathcal{L}_{recon} = \|\mathbf{C} - \hat{\mathbf{C}}\|_2^2$.

We train all scenes using an NVIDIA GeForce RTX 3090 until the reconstruction loss converges, fluctuating by $\sim 10^{-3}$ between iterations. Further detail concerning the original Plenoxels implementation can be found within the [original paper](#).

3.2. Robust Object Capture

Identifying Objects. Segmentation masks were generated using Segment Anything 2 (SAM2) [RGH*24]. To correct for missed object pixels, like small structures, we slightly expand masks by convolving them with a 4×4 dilation kernel prior to training. This leaves over-estimation as the primary source of error. For efficiency, we cast rays through only true-valued mask pixels. To sharpen object capture, we apply a softplus remapping to density: $\sigma' = \text{softplus}(\alpha\sigma + \beta)$. During training, α and β gradually increase from 1 to ~ 8 and from 0 to ~ 2 , respectively. The latter shifts the softplus transition zone, boosting gradients for low-density regions while the former sharpens the transition from “empty” to “solid”. In our experiments, this process can make it easier to isolate confident structure as a prominent, non-zero peak in density is induced, as illustrated in Fig. 3, which can be used to guide thresholding.

Priors. Even with perfect masks, rendering only foreground objects is an ill-posed problem with degenerate solutions. To mitigate this, we obtained depth estimates, $\hat{\mathbf{D}}$, via Depth Pro [BDG*24] for all but one scene, which used Depth Anything V2 [YKH*24]. A Pearson correlation (ρ) loss then guides rendered depth, $\mathbf{D}$, as it is cheap to compute as well as shift and scale invariant: $\mathcal{L}_{depth} = \lambda(1 - \rho(\mathbf{D}, \hat{\mathbf{D}}))$. We decrease λ from 8×10^{-2} to 1.6×10^{-2} during training. While depth estimates provide a geometric prior, rendering only targets prevents supervision of “incorrect” rays, allowing the model to “cheat” depth integrals by distributing false density along the ray, resulting in physically inaccurate density that still obeys correct ordering. An example of such cheating is shown in

Fig. 5. This can be overcome by imposing a minimum distance to density: we block density between the camera and a visual hull, found in the standard way by projecting voxels into image space across views and culling based on consistent alignment with masks. While visual hull-guided sampling has been employed in isolation, they are a coarse technique; as maximally consistent volumes, visual hulls are known to over-estimate [Lau94], particularly in the case of concave structures. In practice, such visual hull errors may be compounded by poor masks and incomplete viewpoints. Depth information is used to help to reduce degeneracy and prevent erroneous background pixels from clouding the visual hull. Together, these two strategies provide a robust set of reconstruction constraints. The dilation described in the previous section also exposes a very thin boundary around objects, revealing depth discontinuity. **Leveraging Failure.** In our experiments, segmentation over-estimation often stems from thin structures that are hard to segment. Such regions tend to carry high semantic value despite occupying minimal space, making them especially important to prioritise when sampling is expensive. To bias sampling towards these areas, we subtract refined masks from coarse ones and clip values < 0 , locating over-estimation. We then dilate the result to expand into nearby structure and reapply the refined mask. This isolates regions of "true" structure that are bounded by SAM2 over-estimation.

4. Results and Evaluation

4.1. Data

Our methodology is evaluated on a [set of object-centric scenes](#) which we name "Plant", "Flower", "Fence", and "Dino Skull" according to their contents. These scenes consists of 52-65 training images each, rendered using the 3D creation software Blender [Ble25]. The public dataset also includes camera intrinsics, camera-to-world poses, coarse segmentation obtained directly from SAM2, and ground-truth segmentation generated using source geometry. As each object possesses some form of thin structure, many SAM2 masks contain over- or under-estimation errors, examples of which are shown in Fig. 1. While real data would be preferable, this inherent difficulty motivates the use of synthetic data, allowing us to quantitatively evaluate reconstruction robustness using ground-truth masks which would not be possible using real scenes. Ultimately, this dataset serves as a specialized benchmark that enables researchers to assess the robustness of object-centric reconstruction techniques or build mask refinement strategies.

4.2. Reconstruction

After training, we remove mask over-estimation by sampling the scene and counting pixel rays that intersect with dense voxels. To evaluate reconstruction, we compute the intersection of union (IoU) between these refined masks and ground truth. IoU serves here as a proxy measure of geometry recovery, not a segmentation benchmark - results are shown in Tab. 1. Improvements over SAM2 show that our method produces meaningful geometry and reduces the influence of erroneous background pixels. While average IoU gains for Dino Skull and Flower were marginal – less than 0.01 for the former - these scenes contained more subtle segmentation error. Additionally, the point clouds (produced by thresholding dense

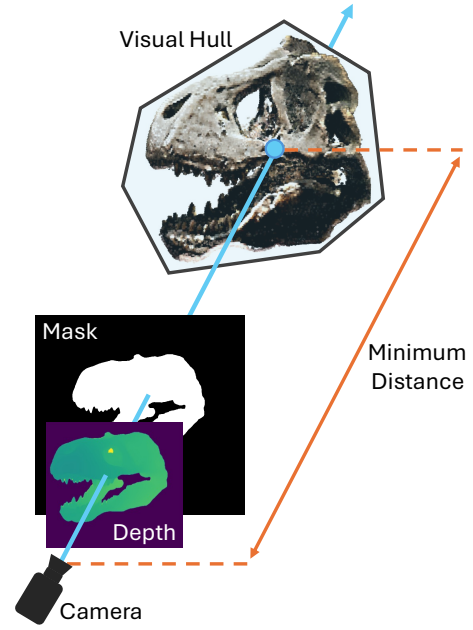


Figure 2: A graphical overview of the key steps in our reconstruction pipeline.

Table 1: Average IoUs between ground truth masks and those predicted by SAM2 as well as those generated using dense voxels within our representation.

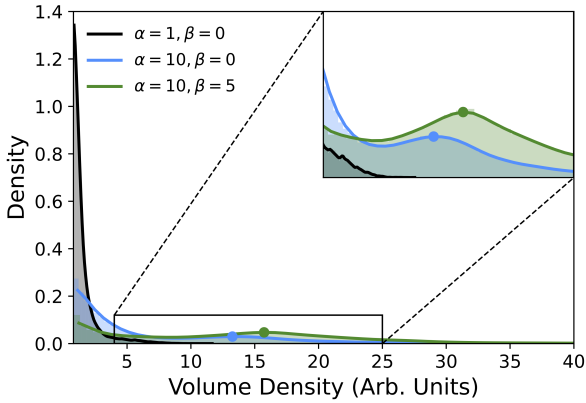
	IoU $\uparrow$	
	SAM2	Ours
Plant	0.78	0.84
Flower	0.94	0.95
Fence	0.79	0.84
Dino Skull	0.95	0.95

voxels) shown in Fig 4 demonstrate significant improvement over the baseline, which includes no visual hull or depth information. Benefits are visible for all four scenes, recovering fine structure despite the types of error exemplified in Fig. 1.

Tab. 2 quantitatively corroborates this advantage with average Structural Similarity Index (SSIM) values obtained from segmented novel views. These were rendered from both the baseline and our full method - in all cases, the baseline renders are inferior. As SSIM aligns with human perception, being sensitive to textures and patterns, these results suggest that our constraints help reconstruct true structure more faithfully. Insight into each constraint's behavior in isolation is provided by Fig. 5; while inflated visual hulls allow density to leak into the object region, depth-only reconstruction appears structurally superior despite inaccurate color. Such inaccuracy is caused by appearance information being distributed along unsupervised rays, as shown in the full scene image. This is in spite of an average correlation of 0.884 with respect to monocular estimates - since depth is an expectation value, there is no hard requirement for a single, solid surface and therefore many possible solutions exist in an object-centric setting.

Table 2: Average SSIM values obtained from segmented novel views rendered from both the baseline and our full model.

	SSIM $\uparrow$	
	Baseline	Ours
Plant	0.84	0.88
Flower	0.87	0.92
Fence	0.87	0.89
Dino Skull	0.86	0.91

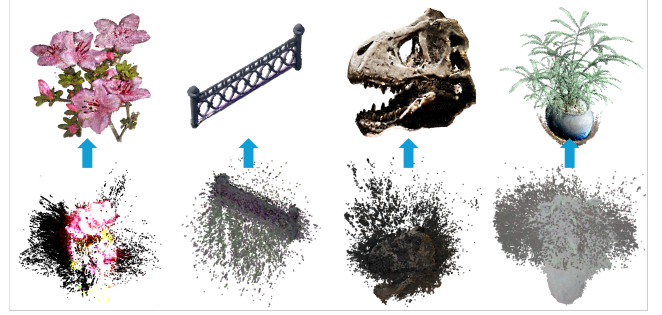
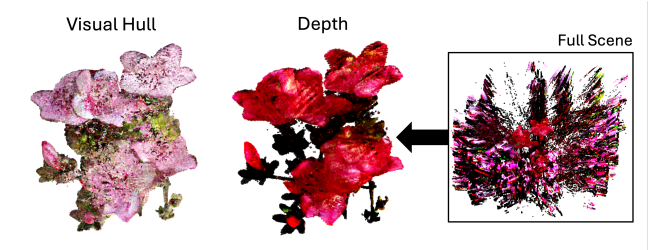
**Figure 3:** The distribution of volume density within the Plant scene for various α and β configurations after 1200 iterations.

4.3. Cues from Segmentation Failure

To demonstrate how refined masks can be used to isolate thin structures, we follow the procedure defined in Section 3.2 for the Plant scene. By detecting over-estimation and expanding it into confident object regions, sampling bias maps are created. Such a map is applied to an example view and shown in Fig. 6. It is clear that fine structure like stems and leaves dominate this bias map while central, dense foliage and the plant pot are neglected - regions that require less dense sampling in inference. While sample points can be concentrated along rays from non-zero map pixels, fine structure could be prioritized in other ways, perhaps by adjusting the underlying grid resolution in these regions.

5. Conclusions and Future Work

We have presented a framework that allows for the robust object-centric capture of 3D assets from poorly segmented images. Together, correlation with monocular depth estimates and visual hull-imposed sampling limits appear to tackle ambiguities inherent to object-centric training as well as those brought by coarse segmentation. This was demonstrated quantitatively with IoU and SSIM improvements as well as visually with point clouds that show significant reconstruction improvements over an under-constrained baseline. Given the goal of forming modular 3D assets, such qualitative results are especially revealing. Furthermore, we illustrate a promising extension, exploiting over-estimated masks to identify fine structure candidates for biased sampling, helping further enhance the capture of thin-structured objects. For practitioners, this

**Figure 4:** RGB point clouds produced by thresholding the baseline method (below) and our pipeline (top).**Figure 5:** A visual ablation for the Flower scene, demonstrating the limitations of each constraint when used in isolation.

framework and our illustration of failure modes can help efficiently build high-fidelity 3D assets from challenging, “in-the-wild” 2D captures. Such volumetric assets may then be converted into standard mesh formats for integration into modern computer graphics pipelines.

Although our results are promising, there are many avenues for further work. Firstly, our method relies on high-quality depth estimation. While we have demonstrated that recent models provide powerful priors in under-constrained regimes, monocular depth estimations rely on 2D, textural cues and can therefore degrade when encountering difficulties like reflections or transparency. In these cases, it would be interesting to probe the limits of depth reliance – perhaps unconfident/erroneous views can be discarded without significantly loosening constraints. Additionally, estimations from alternative methods that consider multiple views, like Structure from Motion, could be used. While we chose a voxel-based method due to its speed, volumetric nature, and history in computer graphics, much of this procedure should be directly applicable to NeRFs, and this would make interesting future work. Furthermore, our dataset can be used to assess the robustness of other representations, like 3DGS. Finally, our demonstration of fine structure identification could be bolstered with further practical experimentation.

6. Acknowledgments

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Figure 6: Predicted fine structure mask applied to a Plant view.

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